Traffic Simulation and Optimization: a comparative study on roundabouts

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Abstract. Urban traffic congestion is a major challenge for modern cities. It is driven by increasing population density and is associated with significant environmental impacts, such as greenhouse gas emissions. These recurrent congestion problems disrupt mobility and highlight the need for efficient traffic-management strategies based on mathematical modeling and data analysis. Although numerous studies have investigated traffic flow, most models focus on individual roads or simplified networks, leaving a gap in modeling at the scale of an entire road system. This project aims to model traffic at the scale of a simple intersection and to apply the approach to a real case study in Germany. The method is based on the Lighthill–Whitham–Richards (LWR) model, a macroscopic framework that describes traffic flow through partial differential equations. Numerical simulations are used to capture traffic dynamics on a network of interconnected roads and to assess which policies can promote smoother and more efficient traffic flow through the optimization of key parameters, such as maximum speed. The results highlight the difficulty of accurately modeling traffic at this scale, mainly because the lack of detailed and reliable data limits calibration and validation. Nevertheless, we were able to study different roundabout layouts and conduct a comparative analysis of the intersection. In conclusion, while large-scale traffic modeling remains challenging, the LWR model provides a solid framework for analyzing traffic flow. Future work should focus on improved data availability and more realistic network representations.

Keywords: LWR model, T-junction, roundabout simulation, multi-class traffic flow, Godunov scheme, Riemann problem, demand–supply method.

Introduction

Urban traffic congestion is a major challenge for contemporary cities, with direct consequences for travel time, fuel consumption, air quality, and the overall quality of life of nearby residents. Because these effects emerge from the collective motion of very large numbers of vehicles, macroscopic traffic models remain particularly attractive for urban analysis: they provide a computationally efficient description of traffic at the scale of roads and networks while preserving the main mechanisms responsible for the onset and propagation of congestion.

Among these models, the classical Lighthill–Whitham–Richards (LWR) scalar conservation law [9, 11] occupies a central place in traffic-flow theory. In its simplest form, it describes the evolution of a density field $\rho(x,t)$ through the conservation equation

$$\partial_t \rho + \partial_x\big(\rho\, v(\rho)\big) = 0 \tag{1}$$

closed by a fundamental diagram relating flow and density. This framework is able to reproduce essential macroscopic phenomena such as shock waves, rarefaction waves, queue formation, and backward propagation of congestion. However, while the single-road setting is by now well understood, extending the model to intersections and road networks is significantly more complex because vehicles may enter, leave, merge, or split according to competing priorities and routing choices. The broader modeling background used in this project is drawn from the overview and pedagogical documents in [6–8].

This difficulty is especially visible at roundabouts and T-junctions, where local interaction rules strongly influence global traffic performance. In that context, Mercier [10] proposed a rigorous framework for a *one-T junction*, viewed as a point where a secondary road connects to a main carriageway. The key idea is to decompose the total density into several vehicle classes, $r = \rho_1 + \rho_2$ upstream and $r = \rho_1 + \rho_3$ downstream, thereby distinguishing through traffic, exiting vehicles, and entering vehicles. Each class satisfies its own conservation law driven by the common total density, while the coupling at the junction is enforced by boundary conditions written as flow inequalities and resolved through a maximisation principle. This formulation provides a mathematically sound description of the Riemann problem at the junction and guarantees the existence and uniqueness of an entropy weak solution under suitable hypotheses on the speed law.

Despite these theoretical advances, the practical exploitation of such results for simulation and comparison of concrete layouts remains only par-

tially developed. The present work therefore aims to bridge the gap between theory and application. More precisely, we implement the multi-class LWR framework numerically, starting from simple benchmark configurations and progressively extending it to a full roundabout model. We then apply this approach to a real case study based on traffic data collected in Germany, with the goal of comparing road layouts and identifying configurations that are less likely to generate persistent congestion and its environmental consequences. Particular attention is paid to the role of entry capacities, priority rules, and routing choices, as well as to the limitations induced by incomplete or weakly saturated data, which make calibration and validation difficult.

The broader project combines a theoretical review of the LWR model and its wave structure with a simulation-oriented study of intersections and roundabouts. In this article, we focus on the numerical component. After recalling the multi-class extension of the LWR model, we study a series of increasingly complex configurations, from a simple road segment and a single T-junction to multi-branch roundabout geometries, before turning to a comparative case study of a real intersection. This progression is meant both to validate the numerical approach on controlled examples and to assess its relevance for more realistic traffic-engineering questions.

1 The Multi-class LWR model

1.1 The LWR Model

The Lighthill-Whitham-Richards (LWR) model represents the foundational framework of macroscopic traffic flow theory [9] [11]. By adopting a continuum mechanics perspective, the model treats vehicular traffic as a homogeneous one-dimensional fluid.

The core of the LWR model is grounded in the Law of Conservation of Vehicles, which stipulates that, for a closed road segment without entries or exits, the variation in the number of vehicles is strictly due to the spatial flux. This is expressed by the first-order partial differential equation (PDE):

$$\frac{\partial r}{\partial t} + \frac{\partial q}{\partial x} = 0$$

where:

- $r(x,t)$ is the traffic density (vehicles per unit length).
- $q(x,t)$ is the traffic flow (vehicles per unit time).

A system defined only by the conservation law would be mathematically underdetermined (one equation for two unknowns, r and q). To avoid this problem, the LWR model introduces a Functional Relationship (Equation of State), known as the Fundamental Diagram:

$$q = f(r) = r \cdot V(r)$$

where $V(r)$ is the speed-law, which represents how drivers adjust their speed based on the concentration of surrounding vehicles.

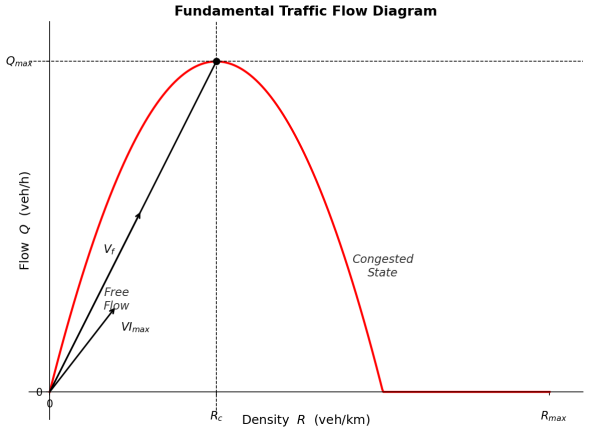


Figure 1: Illustration of a possible fundamental diagram.

As illustrated in Figure 1, traffic dynamics are categorized into three phases. The system remains in a free-flow regime (V_{free}) until reaching the critical density (r_c). Beyond this point, the congested regime begins, where increased density reduces flow.

While the original LWR model assumes a continuous road without interruptions, it has been adapted to model intersections.

1.2 Multi-class LWR model

In the standard LWR model, we consider a single total density r. In the multi-class version, this density is subdivided into specific densities ρ_a, where a represents a specific origin-destination pair.

The theoretical framework presented in this section, follows the approach established by M.Mercier [10]. Unless otherwise stated, the equations and proof structures for the $1-T$ and $n-T$ cases are derived from this work.

The speed law $v(r)$ is shared by all vehicles regardless of their class. Consequently, on any road segment between entries and exits, each class obeys the following conservation law:

$$\partial_t \rho_a + \partial_x(\rho_a v(r)) = 0$$

where r is the density of all classes: $r = \sum \rho_a$.

1.2.1 The T-junction

The basic building block is a road with a single entry/exit point, referred to as a "T-junction". As illustrated in Figure 2.We define three distinct populations:

- ρ_1: Vehicles remaining on the main road.
- ρ_2: Exiting traffic, which exists only upstream of the junction ($x < 0$).
- ρ_3: Entering traffic, which exists only downstream of the junction ($x > 0$).

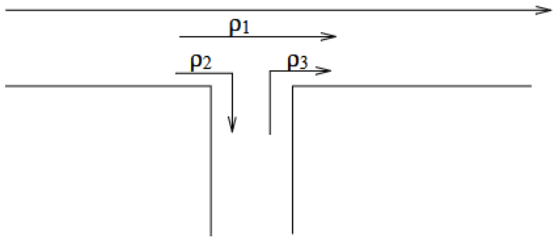


Figure 2: T-junction diagram.

The total density undergoes a change at the junction located at $x = 0$. The local conservation laws are defined as:

$$\begin{cases} \partial_t \rho_1 + \partial_x\big(\rho_1 v(\rho_1+\rho_2)\big) = 0 \\ \partial_t \rho_2 + \partial_x\big(\rho_2 v(\rho_1+\rho_2)\big) = 0 \end{cases} \text{for } x<0, \qquad \begin{cases} \partial_t \rho_1 + \partial_x\big(\rho_1 v(\rho_1+\rho_3)\big) = 0 \\ \partial_t \rho_3 + \partial_x\big(\rho_3 v(\rho_1+\rho_3)\big) = 0 \end{cases} \text{for } x>0. \tag{2}$$

To ensure realism, mathematical solvability and physical consistency we add boundary conditions at $x = 0$:

$$\begin{array}{lcll} \rho_1 v(\rho_1+\rho_2)(t,0-) & = & \rho_1 v(\rho_1+\rho_3)(t,0+) & \max, \\ \rho_2 v(\rho_1+\rho_2)(t,0-) & \leq & o(t) & \max \\ \rho_3 v(\rho_1+\rho_3)(t,0+) & \leq & i(t) & \max, \end{array} \tag{3}$$

The functions $i(t)$ and $o(t)$ represent the respective entry and exit capacities. These conditions ensure that the flow linked to ρ_1 is conserved across the junction, and that the maximum capacity of the road is not exceeded. The "max" notation indicates that we maximize each flow according to a prescribed priority rule. For a roundabout scenario, we prioritize through and exiting traffic (ρ_1, ρ_2); entering traffic (ρ_3) is only permitted to utilize the remaining capacity once the primary demands are satisfied.

To ensure the model is physically consistent, we assume that the speed law $V : [0,1] \to \mathbb{R}_+$ is a strictly decreasing Lipschitz continuous function such that $V(1) = 0$. Furthermore, we assume the flux function $q(r) = rV(r)$ is strictly concave. Under these hypotheses, the Riemann problem defined by the constant initial data:

$$\begin{cases} \rho_1(0,x) = \rho_1^-, \quad \rho_2(0,x) = \rho_2^-, & x<0, \\ \rho_1(0,x) = \rho_1^+, \quad \rho_3(0,x) = \rho_3^+, & x>0. \end{cases} \tag{4}$$

admits a unique weak entropy solution.

The proof involves three fundamental steps: decomposing the junction into two independent 2-population Riemann problems, coupling the upstream and downstream states at $x = 0$, and applying the boundary conditions of Eq. (3) to select the unique solution that maximizes traffic flow under priority constraints. Within the scope of this paper, the theoretical analysis is restricted to the two-population Riemann problem.

1.2.2 Analysis of the Two-Population System

The core of the proof relies on the analysis of a two-population system without junctions. By expressing the original conservation laws in matrix notation, we reformulate the model into its standard conservative form, as shown in the following system:

$$\partial_t U + \partial_x F(U) = 0, \qquad \text{with} \quad U = \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}, \quad F(U) = \begin{pmatrix} \rho_1 V(r) \\ \rho_2 V(r) \end{pmatrix}. \tag{5}$$

where $r = \rho_1 + \rho_2$ is the total density. To solve this, we introduce r and the proportions $s = \frac{\rho_1}{\rho_2}$. The eigenvalues of the system are:

$$\lambda_1(\rho) = V(r) + rV'(r) \quad \text{and} \quad \lambda_2(\rho) = V(r) \tag{6}$$

with corresponding eigenvectors:

$$\vec{w}_1(\rho) = \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}, \quad \vec{w}_2(\rho) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{7}$$

These eigenvectors characterize two distinct wave families:

- **First Family (Genuinely Nonlinear):** Associated with λ_1, this family corresponds to variations where ρ_1 and ρ_2 change proportionally, modifying the total density r. Because $\nabla\lambda_1 \cdot \vec{w}_1 \neq 0$, this field is genuinely nonlinear, producing either *shocks* (when downstream density is higher) or *rarefaction waves* (when traffic accelerates).
- **Second Family (Linearly Degenerate):** Associated with λ_2, this family maintains a constant total density r while the composition s varies. Since $\nabla\lambda_2 \cdot \vec{w}_2 = 0$, the field is linearly degenerate, giving rise to *contact discontinuities* that propagate at the local flow velocity.

By combining these wave families, any jump in initial traffic states can be resolved through a unique sequence of waves. The complete proof can be found in the annex.

1.2.3 The n-T case

The n-T model generalizes the T-junction to a road with n consecutive entry/exit points $x_1 < \cdots < x_n$.

The density $\rho_{i,j}$ denotes vehicles entering at x_i and exiting at x_j, yielding $(n+1)^2$ classes in total. On each interval (x_k, x_{k+1}), all classes share the speed law $V(r)$ and satisfy:

$$\partial_t \rho_{i,j} + \partial_x \Big(\rho_{i,j}\, V\Big(\textstyle\sum_{l,m} \rho_{l,m} \Big) \Big) = 0$$

At each junction x_k, the boundary conditions extend those of the T-junction: through-traffic flows are conserved, while exiting and entering flows are bounded by prescribed capacities $o_k(t)$ and $i_k(t)$, and maximised under the same priority rule (through and exiting traffic take precedence over entering traffic).

The n-T Riemann problem is solved by reducing each junction locally to a one-T structure via three aggregate densities, through, exiting, and entering, and applying the one-T solver. Individual class densities are then recovered using the known origin-destination proportions $p_{i,j}$.

Retaining the assumptions of the T-junction model and further requiring that the origin-destination proportions $\rho_{i,j}$ are known, the Riemann problem admits a unique weak entropy solution on $[0, T]$, with:

$$T \geq \frac{L}{2V_{\max}}$$

where $L = \min_k(x_{k+1} - x_k)$ and $V_{\max} = V(0)$. This bound reflects the finite propagation speed of the model: local solutions at neighbouring junctions cannot interact before time T, allowing them to be consistently patched into a global solution.

This existence result is local in time by construction. Describing the long-time behaviour of the system, in particular on a full roundabout where waves may wrap around and interact, remains an open problem requiring further investigation.

2 Simulation of the LWR model and application to a single-lane roundabout

This section gathers the numerical ingredients and the main simulation outcomes for the single-lane roundabout model. We proceed from the elementary T-junction solver to the full-ring geometry, so that each numerical experiment directly supports the network-scale simulation. The objective is twofold: first, to verify that the discrete scheme reproduces the entropy-consistent wave patterns predicted by the multi-class LWR theory; second, to show that the same solver captures practically relevant roundabout phenomena such as priority-induced metering, queue spillback, and capacity drop.

2.1 Numerical discretisation and junction treatment

The numerical resolution relies on a first-order Godunov finite-volume scheme. Each road of length L is partitioned into N_x cells of width $\Delta x = L/N_x$, and the time step is chosen to satisfy the CFL condition $\Delta t \leq 0.5\, \Delta x / V_m$. At each interface, the total flux is computed through the demand–supply formulation

$$F_{i+1/2} = \min\big(D(r_i), S(r_{i+1})\big), \tag{8}$$

with Greenshields demand and supply functions

$$D(r) = \begin{cases} q(r), & r \leq r_c, \\ q_c, & r > r_c, \end{cases} \qquad S(r) = \begin{cases} q_c, & r \leq r_c, \\ q(r), & r > r_c. \end{cases} \tag{9}$$

This choice preserves the correct free-flow/congested transition and provides an entropy-consistent discretisation of the scalar flux

$$F(r) = r\, v(r). \tag{10}$$

For the multi-class system, all vehicle classes share the same speed law and therefore the same total interface flux. The classwise flux is obtained by upwind splitting,

$$F^{(k)}_{i+1/2} = F_{i+1/2} \frac{\rho^{(k)}_i}{r_i}, \qquad k \in \{1, 2, 3\}, \tag{11}$$

which is consistent with the downstream transport of class proportions. At the junction itself, the standard Godunov flux is replaced by a constrained maximisation that enforces conservation of the through flow while imposing the exit-capacity bound $o(t)$ and the entry-capacity bound $i(t)$. In practice, the upstream demand is first decomposed into partial demands for through-traffic and exiting traffic; the exit class is capped by the prescribed capacity $o(t)$; any remaining downstream supply is then allocated to the entering class, up to the entry capacity $i(t)$. This procedure enforces the roundabout priority rule in a numerically robust way: circulating vehicles are served first, and entering vehicles use only the residual capacity.

2.2 Progressive validation from a single junction to a full ring

We validated the solver through a sequence of increasingly structured test cases, each designed to isolate one feature of the model before moving to the complete roundabout.

Stage 1: scalar sanity check. As a preliminary validation, we simulate a single 1-D road of length $L = 1000$ m with $N_x = 100$ cells and a scalar LWR model. At the midpoint junction, a fraction $\beta = 0.4$ of the arriving flow is removed. The downstream

density drops after the outlet, while a backward-propagating rarefaction develops upstream. This first test confirms that the discrete flux correctly transmits information in both the free-flow and congested regimes.

Stage 2: half-Riemann problems. The second validation stage examines the two half-Riemann problems separately within the full multi-class framework, using the same notation as in Mercier's analysis. For the left half-problem, we denote the upstream state by (ρ_1^-, ρ_2^-) and the associated total density by $r^- = \rho_1^- + \rho_2^-$. In this configuration, the exiting class ρ_2 is constrained by the prescribed exit capacity. The exit bottleneck becomes active when the admissible outflow is smaller than the demand of the exiting class, that is,

$$o < \rho_2^- \, v(r^-), \qquad r^- = \rho_1^- + \rho_2^-. \tag{12}$$

Under this condition, vehicles intending to leave cannot be discharged at the required rate, the total density increases in the vicinity of the junction, and a backward-propagating shock forms on $x < 0$. This is precisely the regime displayed in Figure 3. For the right half-problem, we denote the downstream state by (ρ_1^+, ρ_3^+) and $r^+ = \rho_1^+ + \rho_3^+$. The entering class ρ_3 is admitted only after the demand of the through-traffic class ρ_1 has been satisfied, in agreement with the priority rule built into the junction solver. Whenever the entry constraint becomes active, the downstream profile steepens near the interface; the corresponding numerical configuration is shown in Figure 4.

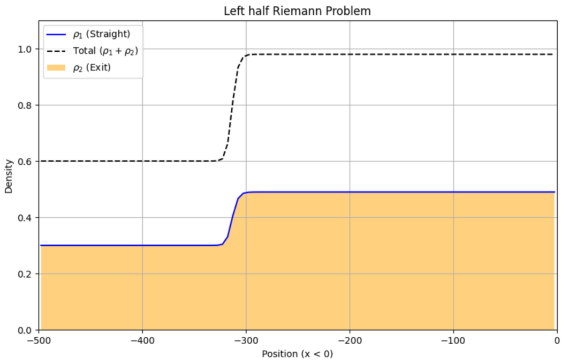


Figure 3: Left half-Riemann problem ($x < 0$), showing the compression of the exiting class and the associated shock structure near the junction.

These separate tests are then combined into a complete T-junction simulation. The representative snapshot shown in Figure 5 exhibits the non-classical discontinuity at $x = 0$ together with the interaction between entering, exiting, and through-traffic classes. The observed density jump is fully consistent with the theory: fluxes remain constrained and conserved according to the admissibility conditions, while the densities themselves may display sharp discontinuities at the junction.

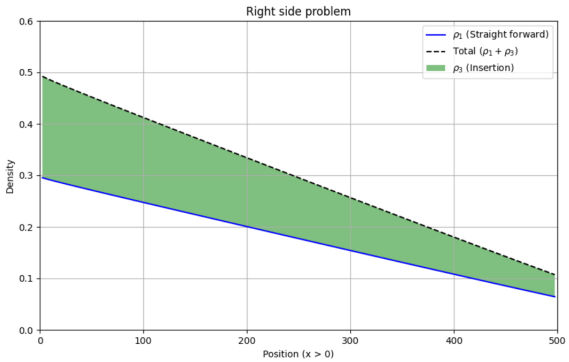


Figure 4: Right half-Riemann problem ($x > 0$), illustrating how entering traffic is admitted only after the through flow has been served.

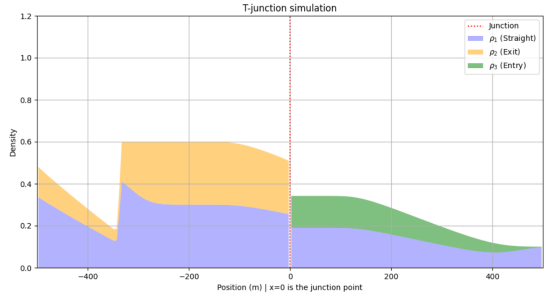


Figure 5: Representative snapshot of the T-junction Riemann problem, showing the interaction of through-traffic, exiting vehicles, and entering vehicles under the roundabout priority rule.

Stage 3: full single-ring roundabout. Having validated the local solver, we assemble a full single-lane roundabout as a circular road of circumference $L_{\text{ring}} = 800$ m, discretised into $N_x^{\text{ring}} = 120$ cells and coupled to three T-junctions J_0, J_1, and J_2. Each junction is connected to one external entry branch, one external exit branch, and, in the extended scenario, to inter-roundabout connector roads. The state vector tracks the six origin–destination classes (i, j) with $i \neq j$, and each local solver reconstructs the aggregates ρ_1, ρ_2, and ρ_3 before applying the constrained maximisation.

Figure 6 summarises the geometry of the computational domain. The periodic closure of the ring makes it possible to follow congestion waves over several turns and to study how local bottlenecks redistribute density over the entire roundabout.

We consider the configuration shown in Figure 6, consisting of six class-dependent fluxes and three T-junctions connected through periodic boundary conditions. The corresponding numerical solution, displayed in Figure 7, is first represented in a one-dimensional unwrapped form, which is the natural output of the finite-volume solver and highlights the

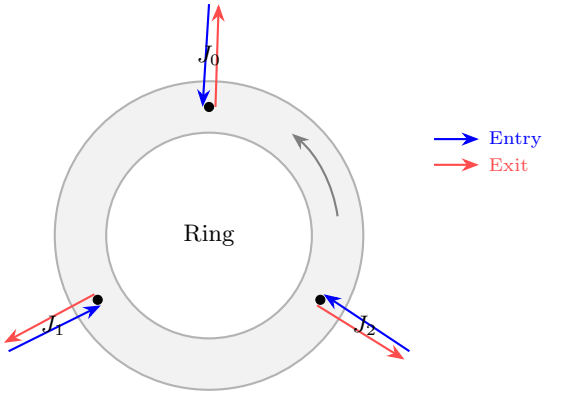


Figure 6: Geometry of the single-ring roundabout with three T-junctions J_0, J_1, and J_2. Blue arrows denote external entries and red arrows denote exits; traffic circulates counter-clockwise on the ring.

evolution of the total density along the ring. The same state is then visualised in Figure 8 through a two-dimensional rendering of the roundabout and its access branches, which makes the spatial localisation of congestion more immediately readable.

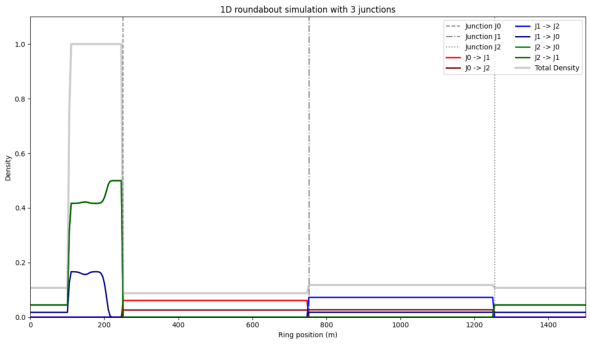


Figure 7: 1-D graph of the roundabout, obtained by unwrapping the ring and plotting the density field as a function of arc-length.

Stage 4: three interconnected roundabouts. To assess scalability, we finally couple three copies of the Stage 3 roundabout and arrange them on a triangular network. Each ring still contains three local T-junctions, but the links between neighbouring rings are now treated as additional roads evolved with the same Godunov finite-volume update. In this setting, congestion is no longer purely local: a queue formed at one entry may fill a connector, reduce the receiving supply of the next ring, and create a secondary queue on a neighbouring roundabout.

In the reported configuration, the complete network contains 600 cells and 90 class-road state variables. The purpose of this last test is therefore to check that the local priority rule remains stable when disturbances circulate through several coupled roundabouts rather than one isolated ring. Figure 9 is reserved for a global screenshot of this network. The interactive dashboard exposes the local parameters (d_k, i_k, o_k, p_k) and lets us observe in real time

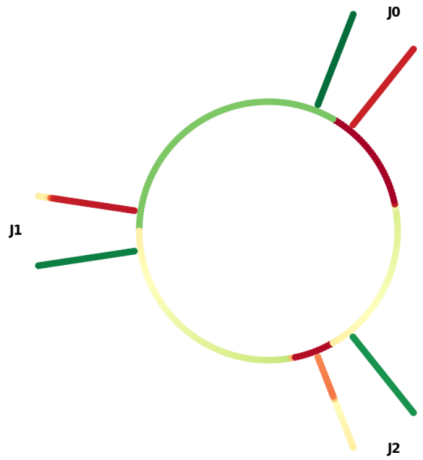


Figure 8: 2-D visualisation of the roundabout and its access branches, highlighting the spatial organisation of congestion around the junctions (red = congested, green = free).

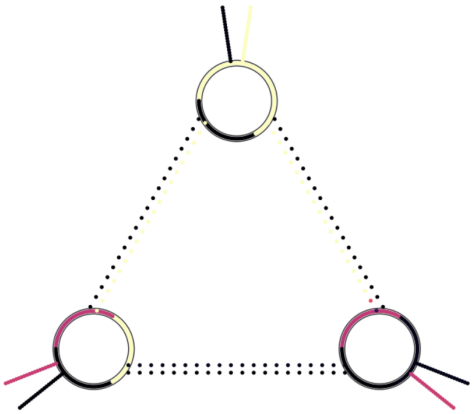


Figure 9: Global view of the three interconnected roundabouts used in Stage 4, colored by density of vehicules in a congestion situation

whether a perturbation remains confined to one node or propagates through the entire three-roundabout system.

2.3 Interpretation, limitations, and outlook

Taken together, these simulations show that the Godunov–demand–supply discretisation reproduces the qualitative wave structure predicted by Mercier's T-junction theory while remaining computationally lightweight enough for network-scale experiments. In particular, the numerical priority rule prevents entering traffic from overwhelming the ring under high external demand and naturally generates queue spillback when downstream supply becomes insufficient.

Several limitations nevertheless remain. First, the

solver inherits the singular behaviour identified in Proposition 3.8 of [10]: when $o \to 0$ and $\rho_2 \to 0$ simultaneously, the numerical solution may exhibit sharp and physically implausible spikes, so small guard thresholds are required in practice. Second, the current scheme is first-order accurate in both space and time; higher-order reconstructions would reduce numerical diffusion near shocks. Third, the capacities i_k and o_k, together with V_m and $\rho_{\max}$, still need calibration against measured data. This calibration stage will support the next step of the project, namely a quantitative comparison between roundabout configurations and signalised intersections under matched demand scenarios.

The present section therefore provides the simulation backbone of the study: it links the analytical T-junction model to concrete computations and establishes a validated single-lane roundabout solver before the discussion of more elaborate roundabout geometries in the next section.

3 Macroscopic Simulation of a Multi-Lane Turbo Roundabout

Context

Conventional multi-lane roundabouts raise significant capacity and safety issues, caused by improper lane changes from drivers, creating more conflict points and slowing down the roundabout general flow. In 1996, Dr. Bertus Fortuijn, a Dutch traffic engineer and researcher developed the concept of turbo roundabouts. This innovative design uses physical lane separation forcing drivers to choose their direction before entering the roundabout, which contributes reducing conflict points and improves safety and traffic flow.

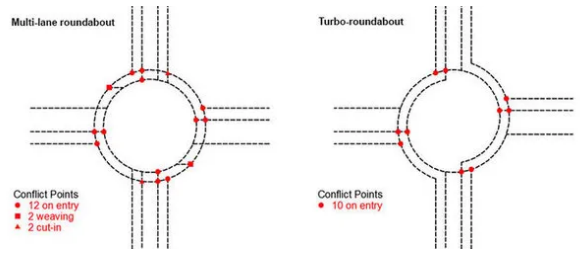


Figure 10: Multi-lane and turbo roundabout designs with visual conflict points (source: turboroundabout.com)

Objective

In this section, we aim to conduct a comparative analysis of the performance of turbo and double-lane conventional roundabouts in order to understand under which conditions it would be relevant for cities and mobility authorities to adopt one design over the other. This work focuses on understanding how local interaction rules (entry capacity, conflicts, lane discipline) influence global traffic behavior (queues, delays, congestion propagation).

After simulating single-lane roundabouts in the previous section, we now turn to double-lane roundabouts. In particular, our objectives are to:

1. Simulate traffic dynamics using the Cell Transmission Model (CTM)
2. Model the entry behavior at the roundabout using gap-acceptance theory
3. Simulate a double-lane roundabout
4. Compare the performance of a double-lane roundabout and a turbo roundabout under identical demand conditions

3.1 Cell Transmission Model (CTM)

We will start by simulating a simple approach road that gets to the roundabout. To simulate the system numerically, we discretize space and time using the Cell Transmission Model. The simulated road is divided into a finite number of cells (in our case 5), and the density is updated at each time step using the following formula:

$$\rho_i^{n+1} = \rho_i^n + \frac{\Delta t}{L}\left(F_{i-1,i}^n - F_{i,i+1}^n\right)$$

where L is the cell length, Δt is the time step, and $F_{i,j}$ denotes the flow from cell i to cell j.

This update formula shows that the density variation in a given cell is determined by the difference between the incoming and outgoing flows.

The flows between cells are then computed using the standard CTM rule:

$$F_{i,i+1} = \min(S_i, R_{i+1})$$

This ensures that the flow is limited both by the ability of the upstream cell to send vehicles and the ability of the downstream cell to receive them.

The sending and receiving functions are defined as:

$$S_i = \min(v_f \rho_i, q_{\max})$$
$$R_i = \min(w(\rho_{\text{jam}} - \rho_i), q_{\max})$$

The sending function S_i represents the maximum flow that the cell i is able to send based on its current density. It increases linearly with density in free-flow conditions and is bounded by the maximum capacity $q_{\max}$.

The receiving function R_i represents the maximum flow that cell i can accept. It depends on

the available space in the cell, given by $(\rho_{\text{jam}} - \rho_i)$, and decreases as the cell becomes congested, with ω being the congestion wave speed.

The CTM is thus able to reproduce key traffic phenomena such as congestion formation, shock wave propagation, and queue dissipation.

3.2 Modeling the Roundabout Entry

After having modeled traffic dynamics on a single road using CTM, we can now introduce the ring of our roundabout and build an elementary conflicting intersection, where the end of our previous 5-cell road will enter the ring and conflict with a circulating stream that is inside the ring.

This elementary simulation will include: one approach road, one entry lane, one ring cell, one-conflicting-stream capacity.

The density flow will not only depend on the sending and receiving capacity of our cells, but also on the gap-acceptance capacity as two streams will be conflicting.

To do this, we will be using a simplified version of Hagring's capacity formula for the gap-acceptance entry capacity C_e that will indicate us the density of cars that can enter the ring taking into account the conflicting circulating flow [veh/h] Q_c, the critical gap [s] T_c, follow-up time [s] T_f and the minimum headway [s] Δ_t.

Gap-Acceptance Capacity

Entry capacity C_e depends on the circulating flow and driver behavior, with $C_e = f(Q_c, T_c, T_f)$ for the one-conflicting-stream case and with $C_e = f(Q_{ce}, Q_{ci}, T_{ce}, T_{ci}, T_f)$ for the two-conflicting-stream case, where Q_c, Q_{ce}, Q_{ci} are the conflicting circulating flows, T_c is the critical gap and T_f is the follow-up time.

These expressions model the fact that vehicles can only enter the roundabout when sufficiently large gaps are available in the circulating traffic.

Entry Flow Model

The actual entering flow at the junction is given by $F_{\text{entry}} = \min(S_{\text{approach}},\ R_{\text{ring}},\ C_e)$ where S_{approach} is the upstream demand (sending function), R_{ring} is the downstream supply (receiving function) and C_e is the gap-acceptance capacity. This formulation highlights that the entry flow is restricted by three possible bottlenecks: demand, available space in the ring, or conflict with circulating traffic.

3.2.1 Single-lane entry

We simulate a single-lane entry that combines the Cell Transmission Model (CTM) for traffic propagation with gap-acceptance functions.

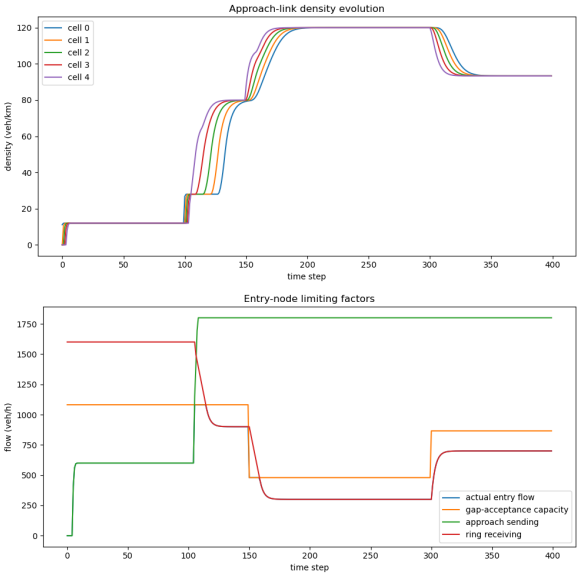


Figure 11: Density evolution on the approach link (top) and entry-node limiting factors (below)

Figure 11 illustrates how congestion on the approach road is directly driven by limiting factors at the entry node.

We observe two main congestion phases. The first congestion begins at around $t \approx 100$, due to a sudden decrease of the ring receiving capacity R_{ring}, caused by an increase of the ring density, which limits the ability of vehicles to enter the roundabout. Concerning the second phase, at around $t \approx 150$ s, congestion intensifies due to an increase in the conflicting circulating flow, which reduces the gap-acceptance capacity C_e. As a result, the entry flow becomes significantly constrained and the queue propagates upstream.

An increase in circulating flow or ring density reduces the entry capacity, which limits the entry flow and generates congestion on the approach road, demonstrating that the entry node is the critical bottleneck of the system. These results put in light the interdependency of the different elements of our system and its direct effect on congestion dynamics.

3.2.2 Double-lane entry

After simulating a single entry lane, we extend the model to a double-lane approach. We consider one approach with two entry lanes (left and right) and a two-lane circulating flow (inner and outer).The two entry lanes face different conflict conditions: the right lane interacts only with the outer circulating lane, while the left lane is affected by the inner and the outer flow, making its entry even more constrained. For simplicity matters, we assume that the left lane only goes to the inner ring. As a result, each lane is modeled independently, with its own density and queue evolution.

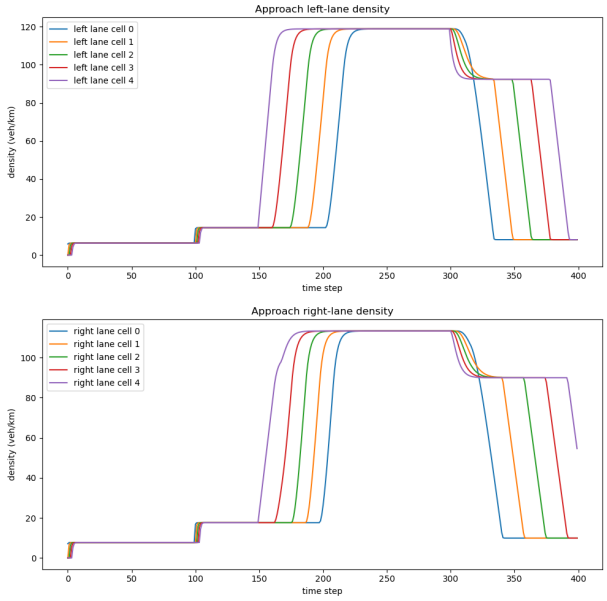


Figure 12: Density evolution on both of the approach road lanes

The two graphs show very similar traffic dynamics and can be divided into four main regimes: a free-flow regime ($t \in [0, 150]$) with stable low densities, a congestion formation phase ($t \in [150, 210]$) where a queue appears at the entry and propagates upstream, a saturated regime ($t \in [210, 300]$) with densities around 110–115 veh/km, and finally a queue dissipation phase ($t \in [300, 380]$) where congestion clears from downstream to upstream.

Even though both graphs display similar behaviors, some differences appear between the two lanes. The left lane reaches congestion earlier and remains congested longer because it is affected by two conflicting streams, leading to a lower entry capacity. In contrast, the right lane, which only faces one conflicting stream, shows a slightly delayed congestion onset and a faster recovery.

These results highlight the impact of the conflict structure on traffic dynamics, with the left lane being more constrained and therefore more prone to earlier and more persistent congestion.

3.3 Turbo and Double-Lane Roundabout Comparison

The construction of the multi-lane roundabout model was carried out progressively through several stages of increasing complexity. We first developed a single CTM link reproducing basic traffic dynamics, before introducing an entry-capacity constraint at a one-entry roundabout. The model was then extended to include two entry lanes and two circulating lanes. This leads us to building a complete four-arm roundabout with dynamic interactions between all approaches in this part.

The final model represents a roundabout with four approaches (North, East, South and West), two entry lanes per approach and two circulating lanes divided into sectors. Conflicting flows are computed at each time step, allowing the model to take the interactions between entering and circulating vehicles into account.

Once the conventional double-lane roundabout model was created, the same framework was adapted to represent a turbo roundabout.

3.3.1 Turbo Roundabout Model

The turbo roundabout model relies on the same macroscopic CTM structure as the conventional double-lane roundabout. However, several important differences appear in the internal traffic logic.

The main distinction concerns lane usage. In a turbo roundabout, lane assignment is predefined, which reduces lane-changing maneuvers and limits interactions between vehicles. In contrast, conventional double-lane roundabouts allow more flexible lane usage.

The entry-capacity formulation also differs between the two configurations. In the conventional double-lane roundabout, the left entry lane interacts with two conflicting streams while the right lane interacts with only one. In the turbo configuration, the capacities depend on whether the approach is considered major or minor, reflecting the more structured organization of traffic streams.

To account for disturbances created by entering vehicles, a merge-friction parameter α is introduced on the outer circulating lane. The effective discharge capacity is therefore written as it follows: $C_{\text{eff}} = C - \alpha F_{\text{entry}}$ where C_{eff} represents the effective circulating capacity, C the nominal capacity and F_{entry} the entering flow. This formulation allows the model to reproduce the slowdown caused by merging interactions.

3.3.2 Total Queue comparison

The two configurations are compared under identical conditions, using the same demand profiles, the same number of lanes and the same CTM discretization structure.

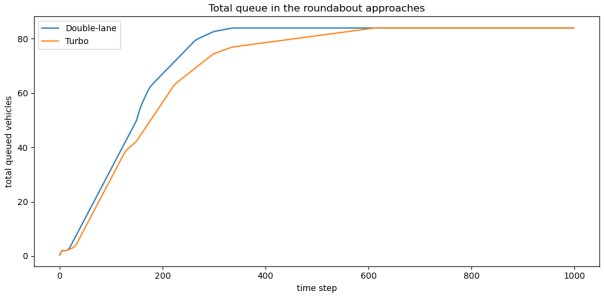


Figure 13: Total queue evolution at every time step for both roundabout approaches

Figure 13 displays two curves representing the total queue of vehicules of the simulation. The total queue increases rapidly during the increasing demand period and eventually reaches the storage capacity of the approach links. The plateau around 84 vehicles corresponds to the maximum number of vehicles that can be stored in the modeled approach cells. The turbo roundabout shows a slower queue accumulation, which indicates a better short-term performance, but it still does not prevent congestion under the simulated scenario. This suggests that the demand level is higher than the effective capacity of both roundabout configurations, for our modeled system.

3.3.3 Density with respect to demand

Traffic conditions can vary significantly depending on the distribution of flows between the main and secondary approaches. To this end, we want to investigate whether the turbo roundabout still outperforms the conventional double-lane roundabout under different traffic demand distributions. To do so, we compute the density difference between the two roundabout configurations for various scaling values of the major and minor-road demands.

This sensitivity analysis helps us identify under which demand conditions the turbo roundabout provides the greatest benefits in terms of queue reduction, with respect to the double-lane conventional roundabout.

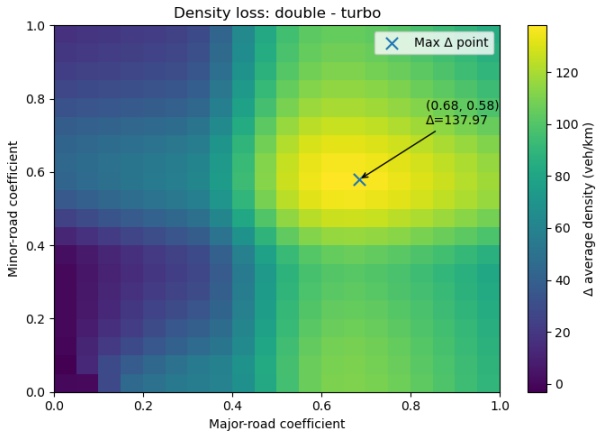


Figure 14: Density loss conventional-turbo roundabouts with respect to major and minor inflow coefficients

Figure 14 represents the average density reduction obtained by the turbo roundabout compared with the conventional double-lane roundabout for different combinations of major-road and minor-road demand coefficients. The plotted quantity is defined as $\Delta\rho = \rho_{\text{double}} - \rho_{\text{turbo}}$, where ρ_{double} and ρ_{turbo} denote the average network densities obtained with the double-lane and turbo configurations respectively.

Positive values indicate that the turbo roundabout outperforms the conventional double-lane roundabout by reducing the average traffic density within the network.

Several important observations can be made from this heatmap. First, for very low traffic demand levels (bottom-left region), the difference between the two configurations remains relatively small. In this regime, both roundabouts are under uncongested conditions, and the geometric advantages of the turbo roundabout have a limited impact on the overall traffic performance.

As the demand increases, the turbo roundabout progressively becomes more advantageous. The largest density reduction is obtained around major_coeff ≈ 0.68 and minor_coeff ≈ 0.58 for which $\Delta\rho \approx 137.97$ veh/km.

This result suggests that the turbo roundabout performs best under intermediate to high traffic conditions. For very high demand levels, the gain no longer increases significantly and even starts to decrease slightly. This behavior indicates that both systems progressively approach saturation. In this regime, storage limitations, ring congestion, and downstream capacity constraints dominate the traffic dynamics, reducing the influence of the roundabout geometry itself.

3.3.4 Further analysis

From the previous analysis, we can see that in this configuration the turbo roundabout still consistently outperforms for all the tested values of inflow. To go further, it would be interesting to test if this results remain robust when we change the other parameters of our model.

In particular, it would be relevant to focus on parameters that represent real-world decision levers, in other words variables that transportation planners or engineers can modify to improve traffic flow.

We have considered the following real levers :

- Roundabout configuration (`mode`): turbo vs. double-lane design.
- Demand-management measures (`major_coeff` and `minor_coeff`): routing strategies, restrictions, or policies affecting traffic distribution across approaches.

To go further, it is also possible to consider:

- Entry capacity (`qmax`): lane width, flaring, or geometric design of the approaches.
- Merge and conflict conditions (`alpha_merge`): quality of lane markings, visibility, and yielding behavior.
- Turn allocation and lane-use rules (`left_turn_share`, `through_share`, and `right_turn_share`): dedicated turning lanes or lane-assignment policies.

3.4 Conclusion

This macroscopic model successfully reproduces the main traffic dynamics observed in multi-lane roundabouts. Even with a simplified framework, the model captures the influence of local interaction mechanisms such as gap acceptance, conflicting flows, lane assignment and merge disturbances on the global behavior of the system.

The comparison between conventional double-lane and turbo roundabouts highlights the importance of lane priorities in reducing congestion. In order to further investigate the applicability of this modeling framework, we now turn to a case study based on real traffic data.

4 Comparative case study of an intersection

We now focus on a case study of a real life intersection, with data collected and put into macroscopic form.

Our aim is to formulate recommendations on the layout of this intersection to prevent congestion of traffic and its consequences on air and noise pollution, and time of travel of the car passengers.

In a first time, we will explain the context and origin of the data we use and we will afterwards fit the different models we explored in this article to this situation and then conclude on the layout

4.1 Data

4.1.1 The inD Dataset

The empirical basis of this case study is the inD dataset, provided by RWTH Aachen University. This dataset was generated using drone-based aerial videography of multiple intersections in Germany, from which individual vehicle trajectories were extracted to provide localization over time [2].

4.1.2 Macroscopic Aggregation Methodology

Because the raw inD dataset provides microscopic data (individual vehicle positions, speeds, and accelerations sampled at 25 Hz), an aggregation step was required to transition to a macroscopic perspective suitable for the LWR framework. We employed a virtual detector approach, placing four detectors perpendicular to each arm of the intersection at a fixed distance from the center.

For a given time interval Δt, the macroscopic variables were derived as follows:

- The traffic flow q [veh/h] was determined by counting the number of vehicle crossings at each detector.

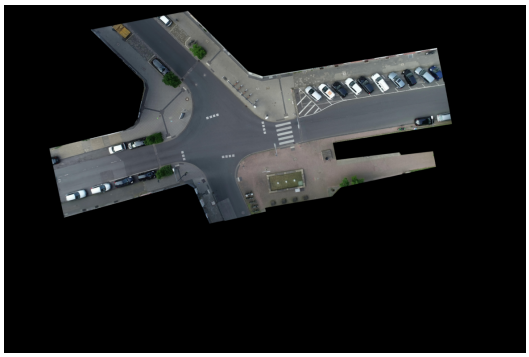


Figure 15: Illustration of the studied intersection

- The space-mean speed v [km/h] was calculated from the instantaneous speeds recorded during these crossing events.
- The traffic density r [veh/km] was derived through the fundamental identity $k = q/v$, ensuring mathematical consistency between the variables and avoiding the geometric complexities inherent in direct spatial vehicle counting.

4.1.3 Site Selection and Observations

The intersection selected for this study was chosen based on two primary criteria: high observed vehicle density and visual clarity of the intersection. We chose the localization presented in the Figure 15

A notable characteristic of the processed data is the absence of saturation. As illustrated in the resulting fundamental diagram in Figure 16, the observations are confined to the free-flow regime, representing only the ascending branch of the expected concave flow-density curve. Consequently, certain LWR model parameters, such as critical density and maximum flow,remain non-identifiable, posing a significant challenge for our simulation. This issue stems from a lack of available data; few datasets are detailed enough, both in terms of time and location to support a thorough analysis. Despite this limitation, Figure 17 allows us to infer a linear decreasing relationship between velocity v and density r.

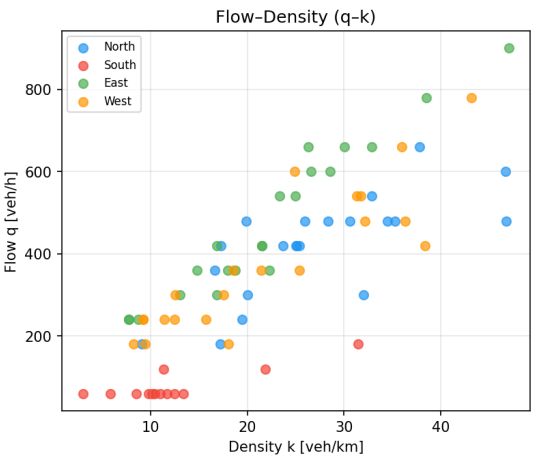


Figure 16: Flow-density diagram

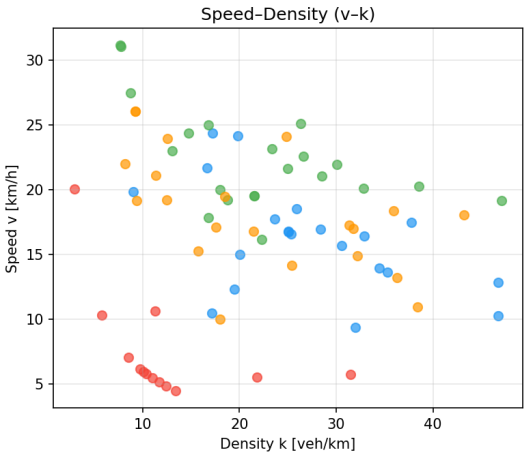


Figure 17: Speed-density diagram

4.2 Model Calibration

The models introduced in the previous sections are generic and must be calibrated before they can be used for a site-specific comparison. In the present case study, calibration consists in identifying the set of parameters that best reproduces both the geometry of the intersection and the macroscopic traffic states extracted from the data. The unknown quantities include physical parameters of the fundamental diagram, such as the free-flow velocity and critical density, together with effective junction parameters describing entry capacities, turning proportions, and priority effects.

Let $\theta \in \Theta$ denote the vector of calibration parameters and let $(v^{\mathrm{obs}}, k^{\mathrm{obs}})$ be the macroscopic observations obtained from the inD dataset. For each admissible vector θ, the numerical solver returns simulated quantities $(v^{\mathrm{sim}}(\theta), k^{\mathrm{sim}}(\theta))$. We then define the error function

$$J(\theta) = w_v \frac{\|v^{\mathrm{sim}}(\theta) - v^{\mathrm{obs}}\|_2^2}{\|v^{\mathrm{obs}}\|_2^2} + w_k \frac{\|k^{\mathrm{sim}}(\theta) - k^{\mathrm{obs}}\|_2^2}{\|k^{\mathrm{obs}}\|_2^2}, \tag{13}$$

where w_v and w_k are positive weights used to balance the relative contribution of speed and density. The calibration problem can therefore be formulated as the inverse problem

$$\theta^\star \in \arg\min_{\theta \in \Theta} J(\theta). \tag{14}$$

This optimisation problem is difficult for several reasons. First, each evaluation of $J(\theta)$ requires a full numerical simulation of the traffic model and is therefore computationally expensive. Second, the map $\theta \mapsto J(\theta)$ is not expected to be smooth, since the Godunov discretisation and the junction rules create regime changes whenever waves interact or capacity constraints become active. Finally, the available observations remain mostly in the free-flow regime, which limits the identifiability of congestion-related parameters. Consequently, the calibrated values should be interpreted as effective parameters for comparative simulation rather than exact physical constants.

4.2.1 Bayesian optimization for model fitting

To address these difficulties, we rely on Bayesian optimization. Following the presentation in Roustant's lecture notes on metamodeling with Gaussian processes [12], Bayesian optimization combines three ingredients: an expensive numerical model, a Gaussian-process surrogate, and a computationally cheap acquisition criterion. The main idea is to replace the objective function J by a probabilistic metamodel that provides, for each candidate parameter vector θ, both a prediction of the objective value and a quantification of the associated uncertainty.

More precisely, if $\mathcal{D}_n = \{(\theta_i, J(\theta_i))\}_{i=1}^n$ denotes the set of previously evaluated points, the surrogate model yields a conditional mean $m_n(\theta)$ and a conditional standard deviation $s_n(\theta)$. The next simulation point is then selected by maximising an acquisition function that balances exploitation and exploration. In a minimisation setting, a standard choice is the expected improvement criterion,

$$\mathrm{EI}_n(\theta) = \mathbb{E}\big[(J_n^{\min} - Y(\theta))_+ \mid \mathcal{D}_n\big], \tag{15}$$

where $J_n^{\min}$ is the best objective value observed after n simulations and $Y(\theta)$ denotes the Gaussian-process predictor at θ. The next evaluation is chosen as

$$\theta_{n+1} \in \arg\max_{\theta \in \Theta} \mathrm{EI}_n(\theta). \tag{16}$$

This criterion favors points located near promising minima while still assigning value to uncertain regions of the parameter space. As a consequence, the simulation budget is concentrated on the most informative experiments.

In the present study, Bayesian optimization is not used with the ambition of recovering a unique and exact set of physical parameters. Rather, it provides an efficient sequential design strategy for exploring a limited number of expensive simulations and identifying parameter values that are consistent with the observed macroscopic behavior. This is particularly important here because the dataset does not contain strongly saturated regimes and therefore cannot fully constrain the descending branch of the fundamental diagram. The resulting calibrated models are thus primarily intended for comparing alternative intersection layouts under a common and coherent calibration framework. In practice, the sequential optimization procedure is implemented with the Optuna library [1].

4.2.2 Practical application

We apply the Bayesian-optimization framework to the different roundabout models introduced in the

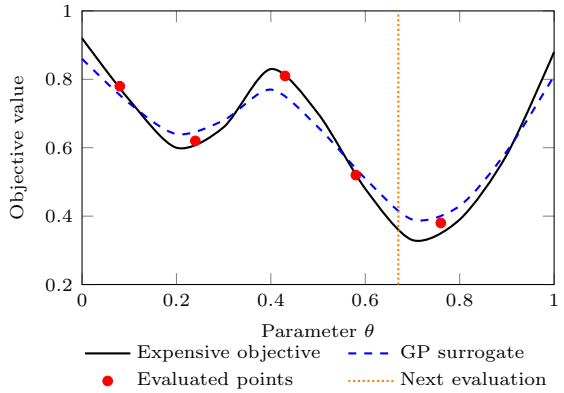


Figure 18: Schematic one-dimensional Bayesian-optimization procedure. After each expensive simulation, a Gaussian-process surrogate is updated and the next evaluation point is selected by maximizing an acquisition criterion such as expected improvement.

previous sections. In practice, the calibration procedure is implemented in Python in a Jupyter notebook, where each traffic model is wrapped as a function of the parameters to be identified. These parameters describe the geometric and operational features of the layout, such as effective capacities, priority-related coefficients, or characteristic values of the fundamental diagram. For any admissible parameter vector, the simulation returns the speed and density values at the same locations as in the real-world observations, which makes it possible to evaluate the discrepancy through the objective function defined above.

The optimization loop is then handled with the Optuna package, which provides a convenient implementation of sequential black-box optimization. In particular, the study is created over a physically meaningful search domain whose bounds are chosen from prior traffic knowledge and from the ranges suggested by the observed data. O

At the end of the procedure, the algorithm returns an estimate of the minimizer of the misfit functional. This estimate is used as the calibrated parameter set for the considered roundabout model and provides the basis for the comparative simulations reported in the next section.

4.3 Comparative study

Although a complete comparative study of the different roundabout layouts cannot yet be presented in this manuscript, the methodological framework required for such an analysis has now been established. The remaining work mainly concerns the completion of the final simulation campaign and the synthesis of the corresponding indicators, which we expect to finalize for the oral presentation.

Once the models have been calibrated, we generate visual animations of the intersection dynamics in order to identify the main congestion zones, queue-propagation patterns, and potential bottlenecks. Starting from this calibrated baseline, we then modify the road-design parameters to evaluate alternative configurations under comparable traffic conditions. Because these parameters retain a clear physical interpretation—for instance, the exit capacity can be related to the width of the outgoing lane—the numerical results can be translated into concrete recommendations for the adaptation of the intersection. This approach also makes it possible to compare the considered layouts on a common basis and to determine whether one configuration is consistently more efficient than the others.

Conclusion

This research highlights the effectiveness of the macroscopic Lighthill-Whitham-Richards (LWR) model in capturing the complex dynamics of traffic at roundabouts. Through the modeling of T-junctions via Riemann problems and the derivation of local-time solutions for n-T junctions using finite volume schemes, the study successfully simulates and visualizes the state of traffic flow within roundabout boundaries. This work confirms that these mathematical frameworks accurately replicate fundamental traffic phenomena like shock waves and queue spillbacks. A primary finding of the comparative analysis is the superior performance of turbo roundabouts over conventional double-lane designs, particularly under intermediate to high demand conditions. This advantage stems from the turbo roundabout's structured lane assignment, which minimizes conflict points and delays compared to the more flexible but conflict-prone conventional layouts. However, the study also reveals that at extreme saturation levels, the geometric benefits of turbo roundabouts diminish as both configurations eventually reach their physical storage and ring density limits. Furthermore, the accuracy of these models remains constrained by the lack of high-resolution, reliable data for precise calibration and validation, as well as certain numerical instabilities that require technical guard thresholds at junction interfaces. Moving forward, future research should focus on integrating more extensive real-world data and implementing higher-order numerical schemes to further enhance the predictive power and practical utility of macroscopic models in urban traffic

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A Annex: Detailed Proof of the Two-Population Riemann Problem

In this section, we provide the detailed mathematical resolution for the two-population system isolated from the junction. We consider the region $x < 0$ with densities ρ_1 and ρ_2, governed by the system:

$$\begin{cases} \partial_t \rho_1 + \partial_x\big(\rho_1 v(\rho_1 + \rho_2)\big) = 0, \\ \partial_t \rho_2 + \partial_x\big(\rho_2 v(\rho_1 + \rho_2)\big) = 0. \end{cases} \tag{17}$$

A.1 Derivation of the Transport Equations

We introduce the total density r and the population ratio s as:

$$r = \rho_1 + \rho_2, \qquad s = \frac{\rho_1}{\rho_2}.$$

By summing the two equations of the system (17), we obtain the conservation of total density:

$$\partial_t \rho_1 + \partial_t \rho_2 + \partial_x(\rho_1 v(r)) + \partial_x(\rho_2 v(r)) = 0.$$

Substituting $r = \rho_1 + \rho_2$, this simplifies to:

$$\partial_t r + \partial_x\big(r\, v(r)\big) = 0.$$

Applying the product rule to the spatial derivative:

$$\partial_x(rv(r)) = v(r)\partial_x r + r\, v'(r)\partial_x r = \big(v(r) + rv'(r)\big)\partial_x r.$$

We thus arrive at the first transport equation:

$$\boxed{\partial_t r + (v(r) + rv'(r))\partial_x r = 0.}$$

Next, we derive the evolution of the ratio s. Applying the quotient rule, we have:

$$\partial_t s = \frac{\rho_2 \partial_t \rho_1 - \rho_1 \partial_t \rho_2}{\rho_2^2}, \qquad \partial_x s = \frac{\rho_2 \partial_x \rho_1 - \rho_1 \partial_x \rho_2}{\rho_2^2}.$$

From the original system, the time derivatives are given by $\partial_t \rho_1 = -\partial_x(\rho_1 v(r))$ and $\partial_t \rho_2 = -\partial_x(\rho_2 v(r))$. Expanding these using the product rule:

$$\partial_t \rho_1 = -v(r)\partial_x \rho_1 - \rho_1 v'(r)\partial_x r, \qquad \partial_t \rho_2 = -v(r)\partial_x \rho_2 - \rho_2 v'(r)\partial_x r.$$

Substituting these expressions into $\partial_t s$:

$$\partial_t s = \frac{\rho_2(-v\partial_x \rho_1 - \rho_1 v'\partial_x r) - \rho_1(-v\partial_x \rho_2 - \rho_2 v'\partial_x r)}{\rho_2^2}.$$

Upon expansion, the terms containing $v'(r)\partial_x r$ cancel out:

$$\partial_t s = \frac{-\rho_2 v\partial_x \rho_1 - \rho_1\rho_2 v'\partial_x r + \rho_1 v\partial_x \rho_2 + \rho_1\rho_2 v'\partial_x r}{\rho_2^2} = \frac{-\rho_2 v\partial_x \rho_1 + \rho_1 v\partial_x \rho_2}{\rho_2^2}.$$

Factoring out $v(r)$, we recognize the expression for $\partial_x s$:

$$\partial_t s = v(r)\left(\frac{-\rho_2 \partial_x \rho_1 + \rho_1 \partial_x \rho_2}{\rho_2^2}\right) = -v(r)\partial_x s.$$

This yields the second transport equation:

$$\boxed{\partial_t s + v(r)\partial_x s = 0.}$$

A.2 Wave Families

The system can be represented in matrix form by introducing the vector $U = (\rho_1, \rho_2)^T$. The conservative form is $\partial_t U + \partial_x F(U) = 0$, with flux:

$$F(U) = \begin{pmatrix} \rho_1 \, v(\rho_1 + \rho_2) \\ \rho_2 \, v(\rho_1 + \rho_2) \end{pmatrix}.$$

Applying the chain rule gives $\partial_t U + A(U)\partial_x U = 0$, where the Jacobian matrix $A(U)$ is:

$$A(U) = \begin{pmatrix} v(r) + \rho_1 v'(r) & \rho_1 v'(r) \\ \rho_2 v'(r) & v(r) + \rho_2 v'(r) \end{pmatrix}.$$

The characteristic polynomial is $\lambda^2 - (2v(r) + v'(r)r)\lambda + v^2(r) + v'(r)v(r)r = 0$, with discriminant $\sqrt{\Delta} = v'(r)r$. The resulting characteristic speeds and eigenvectors are:

$$\lambda_1(\rho) = v(r) + rv'(r), \quad \lambda_2(\rho) = v(r); \qquad v_1(\rho) = \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}, \quad v_2(\rho) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

These define two types of waves:

- **1-Wave Family:** Associated with $v_1(\rho)$, where ρ_1 and ρ_2 vary proportionally. The total density r is modified while the ratio s remains constant (Riemann invariant). Since $d\lambda_1 \cdot v_1 = rq''(r) \neq 0$ (due to strict concavity), this field is *genuinely nonlinear*, producing shocks or rarefaction fans.
- **2-Wave Family:** Associated with $v_2(\rho)$, where r remains constant while s varies. Since $d\lambda_2 \cdot v_2 = 0$, this field is *linearly degenerate*, giving rise to contact discontinuities.

A.3 Construction of the Riemann Solution

For specific initial data cases where $s^- = s^+$:

- If $r^+ < r^-$, a 1-rarefaction wave appears between $\frac{x}{t} = q'(r^-)$ and $\frac{x}{t} = q'(r^+)$.
- If $r^+ > r^-$, a 1-shock wave propagates at speed $c = \frac{r^+ v(r^+) - r^- v(r^-)}{r^+ - r^-}$.

In the general case, the transition from ρ^- to ρ^+ requires an intermediate state ρ_I such that:

1. ρ_I connects to ρ^- via a 1-wave: $s(\rho_I) = s^- \implies \frac{\rho_{1,I}}{\rho_{2,I}} = \frac{\rho_1^-}{\rho_2^-}$.
2. ρ_I connects to ρ^+ via a 2-wave: $r(\rho_I) = r^+ \implies \rho_{1,I} + \rho_{2,I} = r^+$.

Combining these conditions for $r^- \neq 0$, the explicit intermediate state is:

$$\boxed{\rho_I = \frac{r^+}{r^-}\rho^-}$$

This solution confirms that the Riemann problem for the two-population system admits a unique construction based on these characteristic families.